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22MA201
ENGINEERING MATHEMATICS II

UNIT V
COMPLEX FUNCTIONS

1. CONSTRUCTION OF ANALYTIC FUNCTIONS

1.1 MILNE THOMSON METHOD

Let $u(x,y)$ be the real part of the analytic function $f(z)=u(x,y)+iv(x,y)$. In this method, we first find $f^{'}(z)$ as a function of z and then find $f(z)$ by integration. The imaginary part of $f(z)$ gives $v(x,y)$.

Type 1: If real part $u(x,y)$ is given

We construct the analytic function $f(z)=u+iv$ to use the following steps.

(i) Find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$

(ii) Find $\frac{\partial u}{\partial x}(z,0)$ and $\frac{\partial u}{\partial y}(z,0)$

(iii) Using the formula $\int f^{'}(z)dz=\int \frac{\partial u}{\partial x}(z,0)dz-i\int \frac{\partial u}{\partial y}(z,0)dz$ and integrating we get analytic function $f(z)$.

Type 2: If imaginary part $v(x,y)$ is given

We construct the analytic function $f(z) = u + iv$ to use the following steps.

(i) Find $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$

(ii) Find $\frac{\partial v}{\partial y}(z,0)$ and $\frac{\partial v}{\partial x}(z,0)$

(iii) Using the formula $\int f'(z)dz = \int \frac{\partial v}{\partial y}(z,0)dz + i\int \frac{\partial v}{\partial x}(z,0)dz$ and integrating we get analytic function $f(z)$.

1.1.1 APPLICATION OF ANALYTIC FUNCTION:

Fluid flow:

In Fluid Dynamics, the complex potential is a useful quantity used to analyze certain fluid fields. It can be defined by an analytic function

$$f(z) = \phi(x,y) + i\psi(x,y)$$

Where $\phi(x,y)$ is called the **velocity potential.**

$\psi(x,y)$ is called the **stream potential.**

1.1.2 PROBLEMS:

EXAMPLE: 1

An incompressible fluid flowing over the xy-plane has the velocity potential $\phi = x^2 - y^2 + \frac{x}{x^2 + y^2}$. Find a stream function ψ.

Solution:

By Milne – Thomson's method

$$f(z) = \int \phi_x(z,0)dz - i\int \phi_y(z,0)dz$$

Given $\phi = x^2 - y^2 + \dfrac{x}{x^2 + y^2}$.

$$\phi_x = 2x + \frac{y^2 - x^2}{\left(x^2 + y^2\right)^2}, \qquad \phi_x(z,0) = 2z - \frac{1}{z^2}$$

Also,

$$\phi_y = -2y - \frac{-2xy}{\left(x^2 + y^2\right)^2}, \qquad \phi_y(z,0) = 0$$

Therefore,

$$f(z) = \int \phi_x(z,0)\,dz - i\int \phi_y(z,0)\,dz$$

$$f(z) = \int\left(2z - \frac{1}{z^2}\right)dz - i\int 0\,dz = z^2 + \frac{1}{z} + c$$

Now

$$f(z) = (x + iy)^2 + \frac{1}{x + iy} + c$$

$$f(z) = x^2 - y^2 + 2ixy + \frac{x - iy}{x^2 + y^2} + c$$

$$f(z) = \left(x^2 - y^2 + \frac{x}{x^2 + y^2}\right) + i\left(2xy - \frac{y}{x^2 + y^2}\right) + c$$

Hence, the stream function is

$$\psi(x, y) = 2xy - \frac{y}{x^2 + y^2} + c.$$

EXAMPLE: 2

In a two dimensional fluid flow, if $xy\left(x^2 - y^2\right)$ represents the stream function, find the corresponding velocity function and also the complex potential.

Solution:

By Milne – Thomson's method

$$f(z)=\int \psi_y(z,0)dz+i\int \psi_x(z,0)dz$$

Given $\psi = xy\left(x^2-y^2\right)$

$\psi_x = 3x^2y-y^3$, $\qquad \psi_x(z,0)=0$

Also,

$\psi_y = x^3-3xy^2$, $\qquad \psi_y(z,0)=z^3$

Therefore,

$$f(z)=\int \psi_y(z,0)dz+i\int \psi_x(z,0)dz$$

$$f(z)=\int z^3\,dz+i\int 0\,dz=\frac{z^4}{4}+c$$

Now

$$f(z)=\frac{(x+iy)^4}{4}+c$$

$$f(z)=\frac{x^2}{4}+\frac{3x^2y^2}{4}+\frac{y^2}{4}+i\left(x^3y-xy^2\right)+c$$

Hence, the velocity function is

$$\phi(x,y)=\frac{x^2}{4}+\frac{3x^2y^2}{4}+\frac{y^2}{4}+c$$

EXAMPLE: 3

Prove that $u=x^3-3xy^2+3x^2-3y^2+1$ satisfies Laplace equation and determine the corresponding analytic function.

Solution:

Given $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$$u_x = 3x^2 - 3y^2 + 6x, \quad u_{xx} = 6x + 6$$

$$u_y = -6xy - 6y, \quad u_{yy} = -6y - 6$$

$$u_{xx} + u_{yy} = 0$$

Therefore, u is harmonic function.

By Milne – Thomson's method

$$f(z) = \int \phi_x(z,0)\,dz - i\int \phi_y(z,0)\,dz$$

$$u_x(z,0) = 3z^2 + 6z$$

$$u_y(z,0) = 0$$

$$f(z) = \int (3z^2 + 6z)\,dz - i\int 0\,dz$$
$$= z^3 + 3z^2 + c$$

$$u + iv = f(z) = z^3 + 3z^2 + c$$
$$put \quad z = x + iy$$

$$f(z) = (x+iy)^3 + 3(x+iy)^2 + c$$
$$= x^3 + 3x^2 iy + 3(-1)y^2 x + (-i)y^3 + 3x^2 - 3y^2 + 3ixy + c$$

$$u + iv = x^3 - 3y^2 x + 3x^2 - 3y^2 + c + i(3x^2 y - y^3 + 3xy).$$

The harmonic conjugate of u is $(3x^2 y - y^3 + 3xy)$.

EXAMPLE: 4

Prove that the function $u = e^x(x\cos y - y\sin y)$ satisfies Laplace equation and determine the corresponding analytic function $f(z)$ and its harmonic conjugate.

Solution:

Given $$u = e^x(x\cos y - y\sin y)$$

$$\begin{aligned}
u_x &= e^x(x\cos y - y\sin y) + e^x(\cos y)\\
u_{xx} &= e^x(x\cos y - y\sin y) + e^x(\cos y) + e^x(\cos y)\\
&= e^x(x\cos y - y\sin y) + 2e^x(\cos y)\\
u_y &= e^x(-x\sin y - [y\cos y + \sin y])\\
&= -e^x x\sin y - e^x[y\cos y + \sin y]\\
u_{yy} &= -xe^x\cos y - e^x[-y\sin y + \cos y + \cos y]\\
&= -xe^x\cos y + e^x y\sin y - 2e^x\cos y\\
u_{xx} + u_{yy} &= 0
\end{aligned}$$

This implies u satisfies Laplace equation.

By Milne Thomson Method,

$$\begin{aligned}
\int f'(z)dz &= \int u_x(z,0)dz - i\int u_y(z,0)dz\\
u_x(z,0) &= e^z(z(1)-0) + e^z(1)\\
&= e^z(z+1)\\
u_y(z,0) &= e^z(0) - e^z(0)\\
&= 0
\end{aligned}$$

$$\int f'(z)dz = \int e^z(z+1)dz - i\int 0\,dz$$

By using Bernoulli formula, we get $f(z) = (z+1)e^z - e^z + c$.

To find the harmonic conjugate

Put $z = x + iy$

$$
\begin{aligned}
f(z) &= ((x+iy)+1)e^{x}e^{iy} - e^{x}e^{iy} + c \\
&= e^{x}\left[e^{iy}\left[x+iy+1-1\right]\right] + c \\
&= e^{x}\left[\cos y + i\sin y\right]\left[x+iy\right] + c \\
&= e^{x}\left[x\cos y + iy\cos y + ix\sin y - y\sin y\right] \\
u+iv &= xe^{x}\cos y - ye^{x}\sin y + i\left[y\cos y + x\sin y\right]e^{x}
\end{aligned}
$$

The harmonic conjugate of u is $e^{x}\left[y\cos y + x\sin y\right]$.

1.1.3 PRACTICAL PROBLEMS:

1. Construct the analytic function $f(z)$ for which the real part is $e^{x}\cos y$.

   Ans: $f(z) = e^{z} + c$

2. Determine the analytic function $w = u + iv$ if $u = e^{2x}\left(x\cos 2y - y\sin 2y\right)$.

   Ans: $f(z) = ze^{2z} + c$

1.1.4 DISCUSSION QUESTIONS:

1. What is the concept of a complex potential in fluid flow?
2. How are analytic functions utilized in various branches of science and engineering?
3. Application of construction of analytic function in fluid flow?